

Market Competition: Models of Oligopoly

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Static Oligopoly Models

- In a market with a few firms, competition is strategic.
 - firms choose their strategies based on their beliefs (or actions) of their competitors.
- A static oligopoly models:
 - Bertrand model: identical goods, price competition.
 - Cournot model: identical goods, quantity competition.
 - Differentiated goods models.

Bertrand Model

- Two firms: $i = 1, 2$
- Market demand function:
 - $x(p) > 0$ with $x'(p) < 0$.
 - $x(\bar{p}) = 0$ for $\bar{p} > 0$.
 - Constant marginal cost c .
 - $x(c) \in (0, \infty)$.
- The two firms simultaneously choose prices (p_1, p_2) .
- Given (p_1, p_2) firm i captures:

$$x_i(p_i, p_j) = \begin{cases} x(p_i) & \text{if } p_i < p_j \\ \frac{1}{2}x(p_i) & \text{if } p_i = p_j \\ 0 & \text{if } p_i > p_j \end{cases}$$

Nash Equilibrium of the Bertrand Game

- Firm i 's profit is:

$$\Pi(p_i, p_j) = (p_i - c)x_i(p_i, p_j)$$

- (p_1^*, p_2^*) constitutes NE in pure strategies if

$$\Pi_i(p_i^*, p_j^*) \geq \Pi_i(p_i, p_j^*) \quad \forall p_i \geq 0 \text{ and all } i = \{1, 2\}$$

Proposition

$p_1^* = p_2^* = c$ constitutes the unique Nash equilibrium in pure strategies.

Nash Equilibrium of the Bertrand Game

Proof

$p_1^* = p_2^* = c$ is an equilibrium since neither firm can gain from deviation. To show uniqueness, we need to exhaust a few possibilities.

- $p_i^* < c$ for some i .
 i has a profitable deviation to $p_i = c$.
- $p_i^* = c$ and $p_j^* > c$
 i has a profitable deviation to $p_i = p_j - \epsilon > c$.
- $p_i \geq p_j > c$ for all i .
 i has a profitable deviation to $p_i = p_j - \epsilon$.

Cournot Model

- Two firms: $i = 1, 2$
- Inverse market demand function $p(q) = x^{-1}(p)$.
 - $p(q) = a - bq$
- The two firms simultaneously choose quantities (q_1, q_2) and market price is realized $p(q_1 + q_2)$.
- Constant marginal cost.
- Profit function for firm i is:

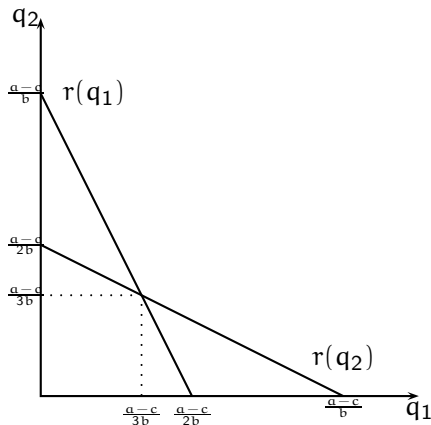
$$\Pi_i = [a - b(q_i + q_j) - c] q_i$$

- Firm i 's best response function:

$$r_i(q_j) = \frac{a - c - bq_j}{2b}$$

Nash Equilibrium

- (q_1^*, q_2^*) constitutes NE if $q_i^* = r_i(q_j^*) \quad \forall i = \{1, 2\}$
- Thus, $q_1^* = q_2^* = \frac{a-c}{3b} > \frac{a-c}{2b}$



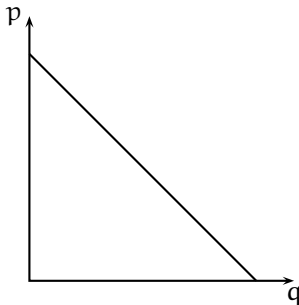
Properties of Cournot equilibrium: Economic Prediction

- $q^* \in (q^c, q^m)$ where q^c and q^m stand for the competitive and monopoly output level.
- $q^*(n)$ is increasing in n where n stand for number of firms.
- $\lim_{n \rightarrow \infty} q(n) = q^c$.
- More realistic prediction than Bertrand competition.
- Unfortunately less realistic situation.
 - most situations involve price competition.

Bertrand Model with Decreasing Returns to Scale

- Let $MC(q_i) = \begin{cases} c & \text{if } q_i \leq \bar{q}_i \\ \infty & \text{if } q_i > \bar{q}_i \end{cases}$
- If $\bar{q}_i + \bar{q}_j < x(c)$, then $p_i^* = p_j^* = c$ is not an equilibrium.
- *Efficient-Rationing Rule*:

$$\tilde{D}(p_j) = \begin{cases} D(p_j) - \bar{q}_i & \text{if } D(p_i) > \bar{q}_i \\ 0 & \text{if otherwise} \end{cases}$$



Bertrand Model with Decreasing Returns to Scale

- Let $q(p) = 1 - p$ (i.e. $a = b = 1$).
- Let $c = 0$.

Proposition

Both firms charging $p^* = 1 - (\bar{q}_i + \bar{q}_j)$ is an equilibrium.

Optimal Capacity Choice

- Let c_0 be per unit cost of building capacity.
- Firms are choosing their capacities simultaneously given $p^* = 1 - (\bar{q}_i + \bar{q}_j)$.

$$\Pi_i^0 = [1 - (\bar{q}_i + \bar{q}_j) - c_0] q_i$$

$$r_i(q_j) = \frac{1 - c_0 - q_j}{2}$$

$$q_i^* = q_j^* = \frac{1 - c_0}{3}$$

- This is precisely the Cournot Output!

Models of Product Differentiation

- If products are differentiated, Bertrand competition does not erode all profits.
- Two classes of models:
 - reduced form: interdependence between the two goods is captured by the two demand functions $x_i(p_i, p_j)$.
 - spacial competition: firms are located apart from each other and it is costly for customers to travel to each location.

Differentiated Duopolists

- $x_i(p_i, p_j) = 1 - p_i + \alpha p_j$
- $c > 0$
 - $\alpha > 0$ -substitutes;
 - $\alpha < 0$ -complements;
- Let $\alpha = \frac{1}{2}$.
- Profit function for firm i is:

$$\Pi_i = \left[1 - p_i + \frac{1}{2}p_j \right] [p_i - c]$$

$$r_i(p_j) = \frac{1+c}{2} + \frac{1}{4}p_j$$

Differentiated Duopolist

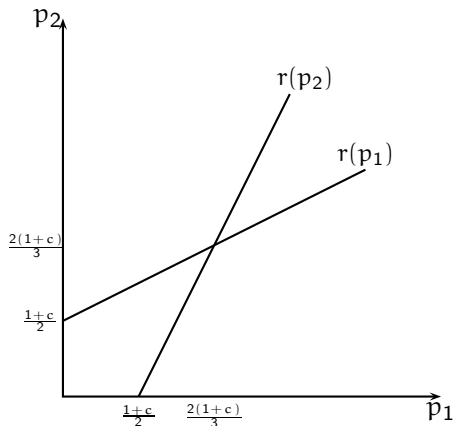


Figure: $p_i^* = \frac{2(1+c)}{3}$

Spatial Competition

- Firms are differentiated either by location or product characteristics.
- 2 Firms: 1 and 2 with constant MC, $c \geq 0$.
- Linear city:

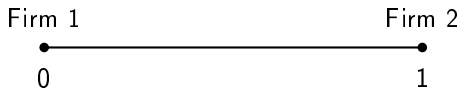


Figure: Consumers uniformly distributed on segment

- Continuum of M consumers uniformly located along the line segment.
 - distance $z \in [0, 1]$ from the origin.
 - gross valuation v for the good.
- Per unit cost of traveling/disutility $t/2$.

Consumer's payoff

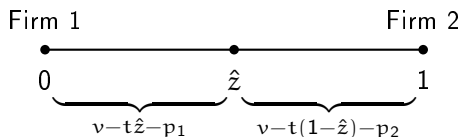


Figure: Consumer located at $\hat{z}$

- Suppose there is an equilibrium, in which the whole market is covered.
- Let consumer at $\hat{z}$ be indifferent between the two goods:

$$v - t\hat{z} - p_1 = v - t(1 - \hat{z}) - p_2$$

$$\hat{z} = \frac{p_2 - p_1 + t}{2t}$$

Equilibrium Derivation

- Given $\hat{z} = \frac{p_2 - p_1 + t}{2t}$, firms' market shares are:

$$F1: M_1 = \frac{p_2 - p_1 + t}{2t} M$$

$$F2: M_2 = \left(1 - \frac{p_2 - p_1 + t}{2t}\right) M$$

- Firm i 's profit function is

$$\Pi_i = (p_i - c)M_i = (p_i - c) \frac{p_j - p_i + t}{2t} M$$

$$r_i(p_j) = \frac{t + p_j + c}{2}$$

Equilibrium Derivation

- Solving for the fixed point:

$$p_i^* = p_j^* = c + t$$

$$M_1 = M_2 = \frac{M}{2}$$

- We need to verify that the market is indeed covered.

$$u(\hat{z}) = v - \frac{t}{2} - (c + t) \geq 0 \implies v \geq c + \frac{3}{2}t$$

References

- Mas-Colell - 12.C
- Tirole J., 1988 "Theory of Industrial Organization", MIT press